

# Hamza Azeem

## TECH LEAD | SOFTWARE ENGINEER

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### PROFESSIONAL EXPERIENCE

#### Amazon | Ads Tech Lead/SWE

##### [Ads Reach Engineering] - Scala • Java • Python • Spark • AWS Dec. 2020 – Present

- Founding engineer and tech lead for a 5-person Reach Engineering team, owning technical direction across campaign forecasting, historical analytics, identity, and data quality for Amazon Ads
- Overhauled the core reach forecasting engine from a one-dimensional sampler to a multi-dimensional supply-matrix and probabilistic sampling architecture, cutting forecast error by more than 50% across core flows and by 80% for guaranteed campaigns
- Re-architected the global reach forecasting platform to automate releases and give each engineer an isolated production-parity stack, cutting time to production from 9-15 days to about 1 hour and local iteration from 7+ minutes to seconds
- Built and scaled historical reach pipelines across 8 marketplaces and 90+ targeting segments; evolved new segment onboarding from 1 week to half a day and cut EMR cost per run by 99% via Spark optimizations and using cost-effective EC2 fleets
- Designed org-wide data integrity and validation platform for 2 PB+ daily across 20+ pipelines; automated executive reporting and reduced release time from days to hours
- Built Amazon Ads' first real-time reach and frequency API, serving sub-200 ms queries on multi-TB Lucene index and powering core Amazon DSP forecasting workflows
- Launched Amazon Ads' first Open Data API, adopted by major global advertising holding companies; led reach measurement for high visibility launches across Prime Video, Twitch, Streaming TV, and other ad programs

##### [Search Data & ML Platform] - Scala • Java • Python • AWS Aug. 2018 – Nov. 2020

- Created Amazon's first low-latency, hourly attributed shopping-behavior dataset, enabling fine-grained model training and analysis affecting millions of customers
- Led the design and delivery of global GDPR automation for Amazon Search, eliminating recurring manual work for thousands of data-subject requests
- Built a centralized marketplace datastore and automated onboarding system, reducing onboarding resource costs by 79% and improving turnaround by 5x
- Reduced schema-change costs on flagship petabyte-scale dataset by 90% by introducing Avro, optimizing MapReduce logic, and migrating the platform to AWS

##### [Visual Search Intern] - Python • Java • R

Aug. 2017 – Dec. 2017

- Developed and deployed the Visual Search team's first machine-learned barcode relevance-ranking models, increasing match rate by 14% and exceeding S-Team target

##### [Visual Search Intern] - Python • AWS

Aug. 2016 – Dec. 2016

- Designed and launched a near real-time ingestion pipeline and API that reduced catalog-update latency from 3 days to under 5 seconds
- Built a backend service supporting near real-time additions, corrections, and deletions in Amazon's barcode-search catalog

### LANGUAGES

- Java
- Python
- Scala
- SQL
- TypeScript

### TECHNOLOGIES

- AWS
  - EMR/Glue/Sfn
  - EC2/IAM/Cfn
  - Lambda/Sagemaker
- Spark
- Lucene
- Git
- \*nix

### EDUCATION

#### University of Waterloo

B.A.Sc in Computer Engineering

2013 – 2018

#### Relevant Courses

- Fuzzy Inferencing Systems & AI
- Distributed Systems
- Operating Systems
- Compilers
- Databases
- Computer Networks

### INTERESTS

- Travel
- Cooking
- Writing
- Fashion